Matrix inequalities and norms

Audenaert's two-row Schatten norm-compression conjecture

Difficulty: extreme **Importance:** broadly interesting

Status: Partially resolved

Rating rationale: The remaining noncommutative norm ranges are a longstanding sharp-inequality barrier; consequences extend to operator geometry and quantum information.

Problem statement

Let $N, r_1, r_2, c_1, \dots, c_N$ be positive integers and let $A_j \in \mathbb{C}^{r_1 \times c_j}$, $B_j \in \mathbb{C}^{r_2 \times c_j}$. For $1 \le p < \infty$, let $\|X\|_p = (\sum_i \sigma_i(X)^p)^{1/p}$, where $\sigma_i(X)$ are the singular values. Define

$$T = \begin{bmatrix} A_1 & \cdots & A_N \\ B_1 & \cdots & B_N \end{bmatrix}, \qquad C_p(T) = \begin{bmatrix} \|A_1\|_p & \cdots & \|A_N\|_p \\ \|B_1\|_p & \cdots & \|B_N\|_p \end{bmatrix}.$$

Do the following inequalities hold for every such partition?

$$\|T\|_p \ge \|C_p(T)\|_p \quad (1 \le p \le 2), \qquad \|T\|_p \le \|C_p(T)\|_p \quad (2 \le p < \infty).$$

Why it matters

The question asks whether the norm of a large block matrix can be controlled sharply using only the norms of its blocks. Such compression estimates are useful when analyzing matrix approximation errors without retaining the full matrices.

References

1. K. M. R. Audenaert, *On a norm compression inequality for* $2 \times N$ *partitioned block matrices*, arXiv:math/0702186v2 (26 March 2007), Conjecture 1, equations (1)–(2), §§2 and 4. Published in Linear Algebra and its Applications 428(4) (2008). Primary text.
2. T. Zhang, *Clarkson–McCarthy type inequalities, part I:* ℓ_p–ℓ_p *and* ℓ_q–ℓ_p *Schatten p-estimates*. arXiv:2410.21961v4 (28 March 2026), Conjecture 1.14 and Theorem 1.15. Primary text.

Status check — 2026-09-10

The 2026 revision still states this conjecture and supplies a weaker bound. Audenaert proves several restricted cases, including $p \ge 4$; the three-row generalization has counterexamples. Hanner's inequality, a special case, has subsequently been proved, which does not settle arbitrary two-row compression. Searches included *Audenaert norm compression conjecture 2026*, *two row Schatten compression proof*, and *norm compression counterexample 2025 2026*; no general resolution was located. Latest arXiv versions were checked for both references.

Audit update (2026-09-10): Rechecked Zhang v4, Conjecture 1.14 and its known-case discussion, and searched for later two-row compression resolutions. The $p \ge 4$ theorem is part of the displayed family; Hanner's theorem does not resolve the other arbitrary-block cases. This is a bounded literature check, not a proof that no solution exists.